

Role of Aerobic Training in Hypertrophy

It is commonly thought that aerobic endurance exercise produces little to no increase in muscle hypertrophy. This belief is consistent with evidence showing that aerobic-type exercise mediates catabolic pathways, whereas anaerobic exercise mediates anabolic pathways. Atherton and colleagues (6) conducted pioneering work to elucidate differences in the intracellular signaling response between the two types of exercises. Using an *ex vivo* model, they electrically stimulated isolated rat muscles with either intermittent high-frequency bursts to simulate resistance-type training or continuous low-frequency activation to simulate aerobic-type training. Postintervention analysis revealed that AMPK phosphorylation in the low-frequency condition increased approximately 2-fold immediately and 3 hours post-stimulation, whereas phosphorylation was suppressed in the high-frequency condition over the same period. Conversely, phosphorylation of Akt was a mirror image of AMPK results: Markedly greater phosphorylation was seen in the high-frequency condition. Recall from chapter 2 that AMPK acts as an energy sensor to turn on catabolic signaling cascades, whereas Akt promotes the intracellular signaling responses associated with anabolism. These findings led to the *AMPK–Akt switch* hypothesis (see figure 6.1), which states that aerobic and anaerobic exercise produces opposing signaling responses and thus are incompatible for optimizing muscular adaptations (6).

Subsequent research, however, indicates that the concept of a switch that regulates anabolic and catabolic signaling pathways is at best overly simplistic and ultimately somewhat misleading. Considerable overlap has been shown to exist between candidate genes involved in aerobic and strength phenotypes, indicating that the two muscle traits are not at opposite ends of the molecular spectrum (70). In fact, multiple studies have shown increased mTOR activation following aerobic endurance exercise (9, 55, 56), whereas resistance training has consistently been found to increase the levels of AMPK (16, 27, 51, 76). To this end, research shows that of 263 genes analyzed in the resting state, only 21 were differentially expressed in aerobic endurance-trained athletes and strength-trained athletes (69).

This chapter discusses how aerobic endurance exercise affects muscle growth. The topic is addressed both when aerobic exercise is performed in isolation and when it is combined with resistance exercise (i.e., concurrent training).

Hypertrophic Effects From Aerobic-Only Training

Contrary to popular belief, a majority of studies show that aerobic training can promote a hypertrophic response in untrained subjects. Reported short-term (12 weeks) gains in skeletal muscle mass from aerobic training are

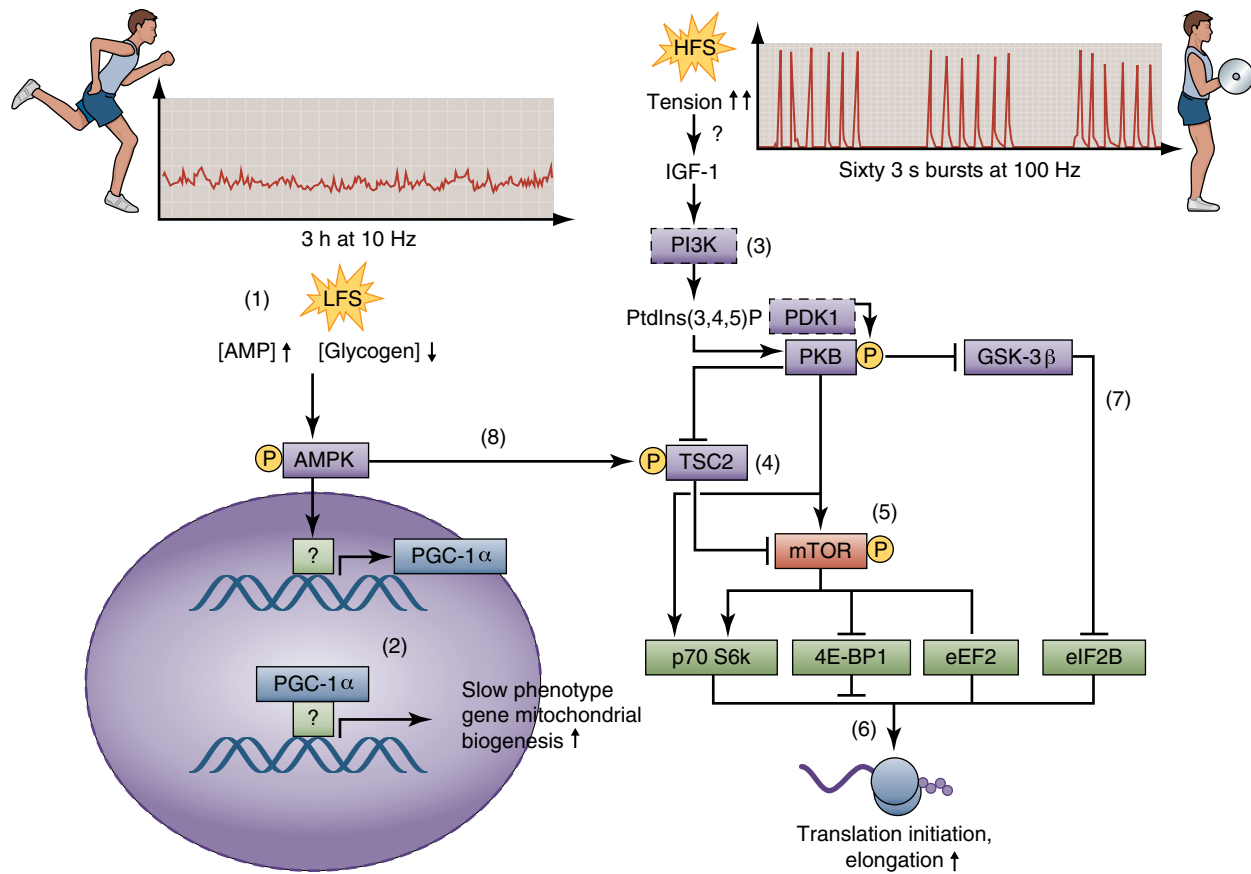


FIGURE 6.1 AMPK–Akt switch hypothesis.

Reprinted by permission P.J. Atherton, J.A. Babraj, K. Smith, J. Singh, M.J. Rennie, and H. Wackerhage, "Selective Activation of AMPK-PGC-1 α or PKB-TSC2-mTOR Signaling Can Explain Specific Adaptive Responses to Endurance or Resistance Training-Like Electrical Muscle Stimulation," *FASEB Journal* 19, no. 7 (2005): 786-788.

similar to those seen in some resistance training protocols, and findings are demonstrated across a spectrum of age ranges in both men and women (50). The following mechanisms have been proposed to account for aerobic exercise-induced muscle growth (50), but the specific roles of these factors and their interactions have yet to be determined:

- Increased insulin-mediated anabolic signaling
- Increased muscle protein synthetic response to nutrition and insulin
- Increased basal postabsorptive muscle protein synthesis
- Increased amino acid delivery
- Increased blood flow and skeletal muscle perfusion

- Decreased myostatin
- Decreased chronic inflammation
- Decreased FOXO signaling
- Decreased protein and DNA damage
- Increased mitochondrial proliferation and dynamics
- Increased mitochondrial energetics (e.g., decreased chronic reactive oxygen species and increased ATP production)

Although most studies have evaluated the muscular adaptations associated with lower-body aerobic training, there is evidence that hypertrophy can be achieved from upper-body arm cycle ergometry as well (74). The extent of hypertrophic adaptations is contingent on intensity, frequency, volume, and mode, in combination with their interaction with genetic

and lifestyle factors. The following sections present the specifics of each of these factors.

Intensity

The body of literature indicates that high intensities are necessary for achieving significant muscle growth from aerobic training. Decreases in muscle cross-sectional area of approximately 20% have been noted in both Type I and Type II fibers after 13 weeks of marathon run training. This indicates that low-intensity exercise is not beneficial to hypertrophy and, in fact, seems to be detrimental when carried out over long durations (72). Although the precise aerobic intensity threshold necessary to elicit hypertrophic adaptations seems to depend on the person's level of conditioning, current research suggests that at least some of the training should be carried out at a minimum of 80% of heart rate reserve (HRR). Training with brief high-intensity intervals (85% of $\dot{V}O_{2peak}$) interspersed with recovery was shown to increase thigh muscle cross-sectional area by 24% in middle-aged people with type 2 diabetes, indicating a potential dose–response relationship between hypertrophy and aerobic intensity, at least in a metabolically compromised population.

Volume and Frequency

Volume and frequency of aerobic training also seem to play a role in the hypertrophic response to aerobic training, a conclusion supported in the literature. Harber and colleagues (38) found that untrained elderly men achieved levels of hypertrophy similar to those of their younger counterparts following 12 weeks of cycle ergometry training despite completing approximately half of the total mechanical workload. These findings indicate that longer periods of sedentarism reduce the total volume necessary for increasing muscle mass, which lends credence to the hypothesis that reviving muscle lost over time is easier to achieve than increasing levels that are close to baseline. Thus, higher aerobic training volumes would seemingly be required in untrained younger people to promote an adaptive response.

The impact of volume may be at least in part frequency dependent. Schwartz and colleagues

KEY POINT

Aerobic exercise can promote increases in muscle hypertrophy in untrained people, but intensity needs to be high—likely 80% of HRR or more.

(63) compared body composition changes in younger versus older men in response to a 6-month aerobic endurance protocol. Each session lasted 45 minutes, and training occurred 5 days per week. Intensity was progressively increased so that participants ultimately worked at 85% of heart rate reserve over the last 2 months of the study. Results showed that only the older men increased muscle mass; no muscular changes were seen in the younger men. The researchers noted that attendance of the younger subjects was significantly less than that of their older counterparts, implying a hypertrophic benefit to greater aerobic training frequency. Notably, it is impossible to tease out the effects of frequency from volume in this study. Whether simply performing longer durations during a single session would confer similar benefits to spreading out frequency over the course of a week has not yet been determined. That said, a hypothetical case can be made that lower-duration, higher-intensity aerobic training performed more frequently would help to optimize hypertrophic adaptations.

Mode

What, if any, impact the modality of aerobic training has on hypertrophic adaptations is unclear. The vast majority of studies on the topic to date have involved cycling exercise, and most of these trials have shown increased muscle protein accretion with consistent training. Studies using noncycling activities have produced mixed results. The previously mentioned study by Schwartz and colleagues (63) found increased muscle mass in elderly but not young male subjects following 6 months of a walk/jog/run protocol. In a study of elderly women, Sipilä and Suominen (66) showed that a combination of step aerobics and track walking at intensities up to 80% of HRR did

not significantly increase muscle cross-sectional area after 18 weeks of training. These findings suggest that it may be more difficult to promote a hypertrophic effect from ambulatory aerobic exercise, perhaps because such activity is performed more often in daily life. Jubrias and colleagues (47) reported no muscle cross-sectional area changes in elderly men and women following a 24-week stair-climbing and kayaking-type aerobic exercise protocol performed with progressively increased intensity up to 85% of HRR.

Other Factors

Although evidence seems to indicate that aerobic training can induce growth in sedentary people, increases in whole-muscle hypertrophy do not necessarily reflect what is occurring at the fiber level. Consistent with its endurance-oriented nature, aerobic-type training appears to produce hypertrophic changes specific to Type I fibers. Harber and colleagues (37) found that Type I cross-sectional area increased by approximately 16% in a group of untrained elderly women following 12 weeks of cycle ergometry training; no change was noted in Type IIa fibers. A follow-up study employing

a similar protocol in younger and older men showed that 12 weeks of cycle ergometry produced an increase in Type I fiber cross-sectional area of approximately 20% (38). Type IIa fiber diameter actually decreased in younger subjects, although not significantly, whereas that of the older subjects remained relatively constant. These findings imply that aerobic exercise may have a detrimental effect on hypertrophy of the faster fiber types. However, other studies show beneficial effects of aerobic training on Type II fiber cross-sectional area in both older (13, 19) and younger (4) subjects. The cause of the discrepancies in findings between studies is not clear.

Evidence also suggests that an increase in mitochondrial proteins is responsible for at least some of the increased fiber growth associated with aerobic endurance training (54). A number of studies have reported that aerobic exercise increases only basal mitochondrial protein synthesis and has no effect on myofibrillar protein synthesis (26, 33, 41, 77). However, work by Di Donato and colleagues (25) showed that both mitochondrial and myofibrillar protein fractions were elevated following an acute bout of high-intensity (90% of maximal

PRACTICAL APPLICATIONS

INTENSITY, FREQUENCY, VOLUME, AND MODE OF AEROBIC TRAINING

Aerobic exercise can increase hypertrophy in sedentary people, primarily in Type I muscle fibers. The extent of hypertrophic increases depends on the level of sedentarism; greater gains are seen in the elderly than in the young. Intensities of $\geq 80\%$ of HRR are generally needed to elicit significant muscular growth. Although definitive evidence regarding the effects of aerobic volume on hypertrophy is lacking, research indicates that longer periods of sedentarism reduce the total weekly duration required to promote the accretion of lean mass. With respect to the modality of exercise, cycling appears to have the greatest hypertrophic benefit, although the paucity of studies on alternative modalities makes it difficult to draw firm conclusions on this variable. Importantly, muscular gains are limited to the early phases after initiating a regimented aerobic exercise program. Results plateau in a relatively short time, and evidence suggests that persistent aerobic training can actually have a detrimental impact on Type II fiber hypertrophy.

heart rate) and low-intensity (66% of maximal heart rate) aerobic exercise. Interestingly, only the high-intensity condition showed sustained muscle protein synthesis elevations at 24 to 28 hours post-exercise recovery. Based on these acute results, it would seem that sarcoplasmic proteins account for a considerable portion of aerobic-induced hypertrophic adaptations. Given evidence that the growth of a given muscle fiber is achieved at the expense of its aerobic endurance capacity (75), the accretion of mitochondrial proteins seems to have a negative impact on the ability to maximize gains in contractile proteins.

An important limitation of current research is that the time course of hypertrophic adaptations during aerobic training has not been well investigated. In those who are sedentary, virtually any training stimulus—including aerobic exercise—is sufficient to overload muscle. This necessarily results in an adaptive response that promotes tissue remodeling. However, the intensity of aerobic training is not sufficient to progressively overload muscle in a manner that promotes further adaptations over time. Thus, it stands to reason that the body would quickly plateau after an initial increase in muscle size.

Early-phase increases in aerobic-induced hypertrophy may be in part due to quantitative or qualitative mitochondrial adaptations, or both. Inactivity induces negative alterations in mitochondrial morphology, and these effects are exacerbated by prolonged sedentarism (15). Mitochondrial dysfunction is associated with increased activation of AMPK and subsequent stimulation of protein degradation, ultimately causing atrophy (34). As previously mentioned, aerobic training enhances the quality and quantity of mitochondrial protein fractions, which would confer a positive effect on anabolic processes. It therefore is conceivable that early-phase hypertrophy in aerobic training is due to restoring normal mitochondrial function and perhaps improving these measures above baseline.

Although aerobic exercise can positively affect muscle mass in the untrained, compelling evidence indicates that it is suboptimal for promoting muscle growth in physically active people. For those who are sedentary, virtually

any stimulus challenges the neuromuscular system and thus leads to an accretion of muscle proteins. Adaptations in these early stages are therefore more indicative of the novelty of the exercise bout as opposed to an increased potential for chronic adaptation. On the other hand, well-trained people have already adapted to lower-level stresses, and it therefore remains highly doubtful that aerobic training would provide enough stimulus for further muscular adaptation. In trained lifters, the mechanical strain associated with aerobic endurance exercise does not rise to the level necessary for mechanotransducers to switch on mTORC1 signaling (76). Indeed, aerobic endurance athletes display slight increases in Type I fiber size while showing a reduction in hypertrophy of Type II fibers (28). Even very intense aerobic exercise does not seem to confer a beneficial hypertrophic effect in those who are highly physically active. This was demonstrated by the finding that 6 weeks of high-intensity interval training resulted in a significant decrease in Type II fiber cross-sectional area in a group of well-trained distance runners (49). Moreover, Mora-Rodriguez and colleagues (59) reported an increased leg fat-free mass in obese middle-aged men following a 4-month aerobic cycling program. However, biopsy analysis revealed that the increases were due to accumulation of intramuscular water; muscle protein concentration actually decreased by 11%.

A recent meta-analysis by Grgic and colleagues (36) compared hypertrophic gains between longitudinal aerobic- versus resistance-exercise programs. Results showed a large difference in effect size (Hedges' $g = 0.66$) for increases in whole-muscle hypertrophy favoring resistance exercise compared to aerobic training. Comparison of individual fiber-type cross-sectional area changes were even more pronounced, with both Type I (Hedges' $g = 0.99$) and Type II (Hedges' $g = 1.44$) fibers displaying very large effects in favor of resistance training. Hypertrophic discrepancies observed between individual fibers and at the whole-muscle level suggest that at least some of the aerobic exercise-induced growth is specific to sarcoplasmic fractions, consistent with acute research on the muscle protein synthetic

response to such training (25). The results held true irrespective of age or sex, further strengthening conclusions. These findings provide compelling evidence that although aerobic training can promote increases in muscle growth, the magnitude of these changes is far inferior to that obtained from resistance training.

In summary, muscular adaptations to aerobic training exist on a continuum, and hypertrophic responses ultimately depend on a variety of individual and environmental factors. Although between-study comparisons suggest that early-phase gains in muscle mass are similar between aerobic and resistance training protocols (31), within-study results indicate a clear hypertrophic advantage to resistance training (see table 6.1). Pooled data from studies directly comparing hypertrophy in the two types of exercise show a strong overall mean effect size difference favoring resistance training both at the whole-muscle and fiber-type level. Moreover, increases in muscle size following aerobic training are not well correlated with increased force capacity, indicating that hypertrophic adaptations are not entirely functional and likely the product of increases in sarcoplasmic protein fractions (54).

Table 6.1 provides a summary of the research comparing the effects of aerobic training versus resistance training on muscle hypertrophy.

Concurrent Training

Aerobic exercise is often performed in combination with resistance training for accelerating fat loss or enhancing sport performance, or both. This strategy, called *concurrent training*, has been shown to have a positive effect on weight management (1). However, evidence suggests that the addition of aerobic exercise to a regimented resistance training program may compromise muscle growth. Negative hypertrophic effects from concurrent training have been attributed to a phenomenon known as *chronic interference* (figure 6.2 on page 161), the hypothesis for which alleges that trained muscle cannot simultaneously adapt optimally morphologically or metabolically to both strength and aerobic endurance training (79). Like the AMPK–Akt switch hypothesis, the chronic interference

hypothesis states that these competing adaptations produce divergent intracellular signaling responses that mitigate muscular gains.

Despite the logical basis for the chronic interference hypothesis, the effect of the phenomenon in humans when performing traditional training protocols is unclear. Although some studies show that combining aerobic and resistance exercise impedes anabolic signaling (17, 18), others have failed to observe any negative consequences (5). There is even evidence that concurrent training heightens mTOR and p70^{S6K} to a greater extent than resistance training alone (53). Moreover, studies show no deleterious effects of concurrent training on muscle protein synthesis (12, 26). Discrepancies in the findings may be related to a number of factors. Importantly, the time course of evaluation in the current literature was generally limited to several hours post-exercise and thus does not provide a complete snapshot of the adaptive response, which can last in excess of 24 hours. Furthermore, these findings are specific to acute bouts of exercise, whereas any interference would seemingly manifest over a period of weeks or months.

It is conceivable that concurrent training negatively affects growth in other ways. For one, acute factors associated with aerobic training may interfere with resistance training capacity. Specifically, aerobic exercise can cause residual fatigue, substrate depletion, or both, which ultimately impairs the quality of the resistance training bout (31). Muscular adaptations are predicated on the capacity to train with an intensity of effort that sufficiently stimulates myofiber growth. If this ability is compromised, muscular gains necessarily suffer.

Another potential issue with concurrent training is an increased potential for overtraining. When the volume or intensity of training exceeds the body's ability to recover, physiological systems are disrupted. The stress of adding aerobic exercise to an intense hypertrophy-oriented resistance training program can overtax recuperative abilities, leading to an overtrained state. The interference effects of aerobic exercise associated with overtraining may be mediated by a catabolic hormonal environment and chronic muscle glycogen depletion (58).

TABLE 6.1 Summary of Research Comparing Hypertrophic Adaptations Between Aerobic and Resistance Training

Study	Subjects	Resistance training protocol	Aerobic training protocol	Aerobic training modality	Study duration; training frequency	Method of hypertrophy assessment (site)	Results
Ahtiainen et al. (3)	19 untrained older men	Periodized total-body, multiset resistance training program varying loads from 40% to 90% of 1RM	Combination of steady-state aerobic exercise and training above the lactate threshold. Duration of steady-state training ranged from 30 to 90 minutes per session; high-intensity intervals ranged from 5 to 10 minutes.	Cycling	21 weeks; twice per week	Ultrasound (vastus lateralis, vastus intermedius), muscle biopsies (vastus lateralis)	Increases in quadriceps muscle thickness and Type II fiber CSA for resistance training; no hypertrophic increases for aerobic training
Bell et al. (8)	22 untrained young men and women	Total-body, multiset resistance training program with loads progressively increased from 72% to 84% of 1RM	Combination of steady-state aerobic exercise and HIIT. Duration of steady state training ranged from 30 to 42 minutes per session; interval training was carried out with a work-to-rest ratio of 3:3 minutes with a duration of 24 to 36 minutes per session.	Cycling	12 weeks; 3 times per week	Muscle biopsy (vastus lateralis)	Increases in Type I and Type II fiber CSA for resistance training; no hypertrophic increases for aerobic training
De Souza et al. (23, 24) *	19 untrained young men	Periodized lower-body, multiset resistance training program varying repetitions from 6RM to 12RM	HIIT carried out at 80% to 95% $\dot{V}O_2$ max with a duration of ~30 to 40 minutes per session.	Running	8 weeks; twice per week	MRI (quadriceps), muscle biopsy (vastus lateralis)	Increases in whole-muscle quadriceps CSA and Type I and Type IIa fiber CSA for resistance training; no hypertrophic increases for aerobic training
Farup et al. (29)	14 untrained young men	Periodized lower-body, multiset program varying repetitions from 4RM to 10RM	Combination of steady-state exercise and interval training. Steady state was performed for 30 to 45 minutes at 60% to 75% of watt max; moderate-intensity interval training was performed for 2 × 20 minutes at 60% to 85% of watt max, with a 5-minute rest interval; HIIT was performed for 8 × 4 minutes at 70% to 90% of watt max with a 1-minute rest interval.	Cycling	10 weeks; 3 times per week	MRI (quadriceps, hamstrings, hip adductors [these components were summed for total thigh CSA]), muscle biopsy (vastus lateralis)	Increases in whole-muscle quadriceps CSA and Type II fiber CSA for resistance training; no hypertrophic increases for aerobic training

Study	Subjects	Resistance training protocol	Aerobic training protocol	Aerobic training modality	Study duration; training frequency	Method of hypertrophy assessment (site)	Results
Ferrara et al. (30)	22 untrained older men	Single-set upper-body resistance training and 2-set lower-body training with loads corresponding to 80% of 1RM	Steady-state aerobic exercise with a duration of 60 minutes per session	Walking or running	36 weeks; 3 times per week	CT (quadriceps)	No increases in muscle CSA for either condition
Hepple et al. (39)	20 untrained older men	Multiset low-body resistance training program with 6 to 12 reps per exercise	Aerobic exercise of unspecified intensity lasting 30 minutes per session	Cycling	9 weeks; 3 times per week	Muscle biopsies (vastus lateralis)	Increases in Type I fiber CSA for resistance training; no hypertrophic increases for aerobic training
Hudelmaier et al. (42)	35 untrained middle-aged women	Periodized low-body, multiset resistance training program varying loading from 60% to 80% of 1RM	Steady-state aerobic exercise with a duration of 60 minutes per session	Cycling	12 weeks; 3 times per week	MRI (quadriceps, hamstrings, hip adductors, sartorius)	Increases in muscle CSA of the quadriceps, hamstrings, adductors, and sartorius for resistance training; increases in quadriceps and sartorius CSA for aerobic training, but no hypertrophic increases in the hamstrings and adductors
Izquierdo et al. (43)	21 untrained older men	Periodized total-body, multiset resistance training program varying loading from 30% to 80% of 1RM	Combination of steady state and training above the lactate threshold. Training sessions lasted for 30 to 40 minutes per session; steady-state training was performed continuously, whereas the training above the lactate threshold had work-to-rest ratios of 30:30 seconds.	Cycling	16 weeks; twice per week	Ultrasound (rectus femoris, vastus lateralis, vastus medialis, vastus intermedius)	Increases in quadriceps CSA for resistance training; no hypertrophic increases for aerobic training

(continued)

Table 6.1 (continued)

Study	Subjects	Resistance training protocol	Aerobic training protocol	Aerobic training modality	Study duration; training frequency	Method of hypertrophy assessment (site)	Results
Izquierdo et al. (44)	21 untrained middle-aged men	Periodized total-body, multiset resistance training program varying loading from 30% to 80% of 1RM	Combination of steady state and training above the lactate threshold. Training sessions lasted for 30 to 40 minutes per session; steady-state training was performed continuously, whereas the training above the lactate threshold had work-to-rest ratios of 30:30 seconds.	Cycling	16 weeks; twice per week	Ultrasound (rectus femoris, vastus lateralis, vastus medialis, vastus intermedius)	Increases in quadriceps CSA for both resistance training and aerobic training
Jubrias et al. (47)	29 untrained older individuals	Periodized total-body, multiset resistance training program varying loading from 60% to 85% of 1RM	Steady-state aerobic exercise lasting 40 minutes per session	One-legged press exercise and a kayaking-type exercise (20 minutes on each mode per session)	24 weeks; 3 times per week	MRI (quadriceps)	Increases in quadriceps CSA for resistance training; no hypertrophic increases for aerobic training
Karavirta et al. (48)	50 untrained older men	Periodized total-body, multiset resistance training program varying loading from 40% to 85% of 1RM	Training at a combination of intensities both below and above the lactate threshold for 30 to 90 minutes per session	Cycling	21 weeks; twice per week	Muscle biopsy (vastus lateralis)	Increases in Type II fiber CSA for resistance training; no hypertrophic increases for aerobic training
Kraemer et al. (52)	17 trained young men	Periodized total-body, multiset resistance training program varying repetitions from 5RM to 15RM	Combination of steady state and training above the lactate threshold. Training sessions for steady state lasted 40 minutes per session; supramaximal aerobic intervals ranged from 200 to 800 seconds per session with a work-to-rest ratio of 1:4 to 1:0.5 minutes.	Running	12 weeks; 4 times per week	Muscle biopsy (vastus lateralis)	Increases in Type I, Type IIa, and Type IIc fiber CSA for resistance training; no hypertrophic increases for aerobic training and decreases in size reported for Type I and Type IIc fibers

Study	Subjects	Resistance training protocol	Aerobic training protocol	Aerobic training modality	Study duration; training frequency	Method of hypertrophy assessment (site)	Results
McCarthy et al. (57)	20 untrained young men	Total-body, multiset resistance training program performed with a repetition range of 5 to 7 per set	Steady-state aerobic exercise lasting 50 minutes per session	Cycling	10 weeks; 3 times per week	CT (quadriceps, flexors), muscle biopsy (vastus lateralis)	Increases in quadriceps CSA for both resistance training and aerobic training; increases in hamstrings CSA in resistance training but no increase in aerobic training; increases in Type I and Type II fiber CSA for resistance training; no hypertrophic increases for aerobic training
Mikkola et al. (58)	27 untrained middle-aged men	Periodized total-body, multiset resistance training program varying loading from 40% to 80% of 1RM	Training at a combination of intensities both below and above the lactate threshold for 30 to 90 minutes per session	Cycling and Nordic walking (one per session)	21 weeks; twice per week	MRI (quadriceps)	Increases in quadriceps CSA for both resistance training and aerobic training
Nelson et al. (60)	9 untrained young men	Lower-body, multiset resistance training program performed 6 repetitions per set	Steady-state aerobic exercise lasting 30 to 60 minutes per session	Cycling	20 weeks; 4 times per week	Muscle biopsy (vastus lateralis)	Increases in Type IIx fiber CSA in resistance training; increases in Type I, Type IIa, and Type IIx fibers in aerobic training
Poehlman et al. (62)	31 untrained young women	Total-body, multiset resistance training program performed at a load of 80% of 1RM	Combination of steady state and training above the lactate threshold. Training sessions for steady state lasted 25 to 45 minutes per session; supramaximal aerobic intervals were performed 4 × 5 minutes at 95% of heart rate maximum with a 3-minute rest.	Walking and running	28 weeks; 3 times per week	CT (quadriceps)	No changes in quadriceps CSA in either condition
Sillanpää et al. (64)	27 untrained older men	Periodized total-body, multiset resistance training program varying loading from 40% to 90% of 1RM	Training at a combination of intensities both below and above the lactate threshold for 30 to 90 minutes per session	Cycling	21 weeks; twice per week	Ultrasound (vastus lateralis, vastus intermedius)	Increases in quadriceps muscle thickness for both resistance training and aerobic training

(continued)

Table 6.1 (continued)

Study	Subjects	Resistance training protocol	Aerobic training protocol	Aerobic training modality	Study duration; training frequency	Method of hypertrophy assessment (site)	Results
Sipilä et al. (67, 68) *	24 untrained older women	Periodized lower-body, multiset resistance training program varying loading from 60% to 75% of 1RM	Steady-state aerobic exercise consisting of walking 1,500 meters to 2,700 meters per session and step aerobics sessions lasting for 40 minutes	Walking (twice per week) and step aerobics (once per week)	16 weeks; 3 times per week	CT (quadriceps, hamstrings, calves), muscle biopsy (vastus lateralis)	Increases in quadriceps and calf muscle CSA, as well as Type I CSA for resistance training; increases in calf muscle CSA for aerobic training, but no other hypertrophic increases noted
Willis et al. (78)	82 untrained older men and women	Total-body, multiset resistance training program performed for 8 to 12 reps per set	Steady-state aerobic exercise lasting ~45 minutes per session	Treadmill (not specified as running or walking), elliptical trainer, cycling	32 weeks; 3 times per week	CT (quadriceps)	Increases in quadriceps CSA for resistance training; no hypertrophic increases for aerobic training

RM = repetition maximum; MRI = magnetic resonance imaging; CT = computed tomography; CSA = cross-sectional area; HIIT = high-intensity interval training.

*Two studies from the same dataset

Adapted from Konopka and Harber (50).

Long-term training studies investigating muscular adaptations to concurrent training have produced conflicting findings. When considering the body of literature as a whole, evidence suggests that aerobic exercise blunts the hypertrophic response to resistance training. A meta-analysis by Wilson and colleagues (79) revealed that effect size for muscular gains was reduced by almost 50% in those who solely lifted weights when aerobic endurance training was added to the mix. However, multiple factors determine how and to what extent aerobic training influences the adaptations associated with resistance training. In particular, the manipulation of aerobic exercise intensity, volume and frequency, mode, and scheduling is paramount in creating the response. The following sections provide an overview of these variables and their effects on resistance training-induced hypertrophy.

Intensity

Research directly assessing the hypertrophy-related effects of aerobic endurance exercise

KEY POINT

Evidence suggests that, over time, aerobic exercise blunts the hypertrophic response to resistance training.

intensities during concurrent training is lacking. Evidence suggests that high-intensity sprint cycle interval training is more detrimental to intracellular anabolic signaling than moderate-intensity steady-state cycling (17, 18). Moreover, the post-endurance-exercise activity of negative regulators of muscle protein synthesis (including AMPK and eIF4EB1) are elevated in an intensity-dependent fashion. In addition, one of the two catalytic isoforms of AMPK (AMPK α 1)—which has been shown to selectively inhibit mTORC1—may be preferentially activated by higher, but not lower, aerobic intensities (31). The apparently greater interference associated with high-intensity training suggests that lower-intensity exercise may be preferable if the goal

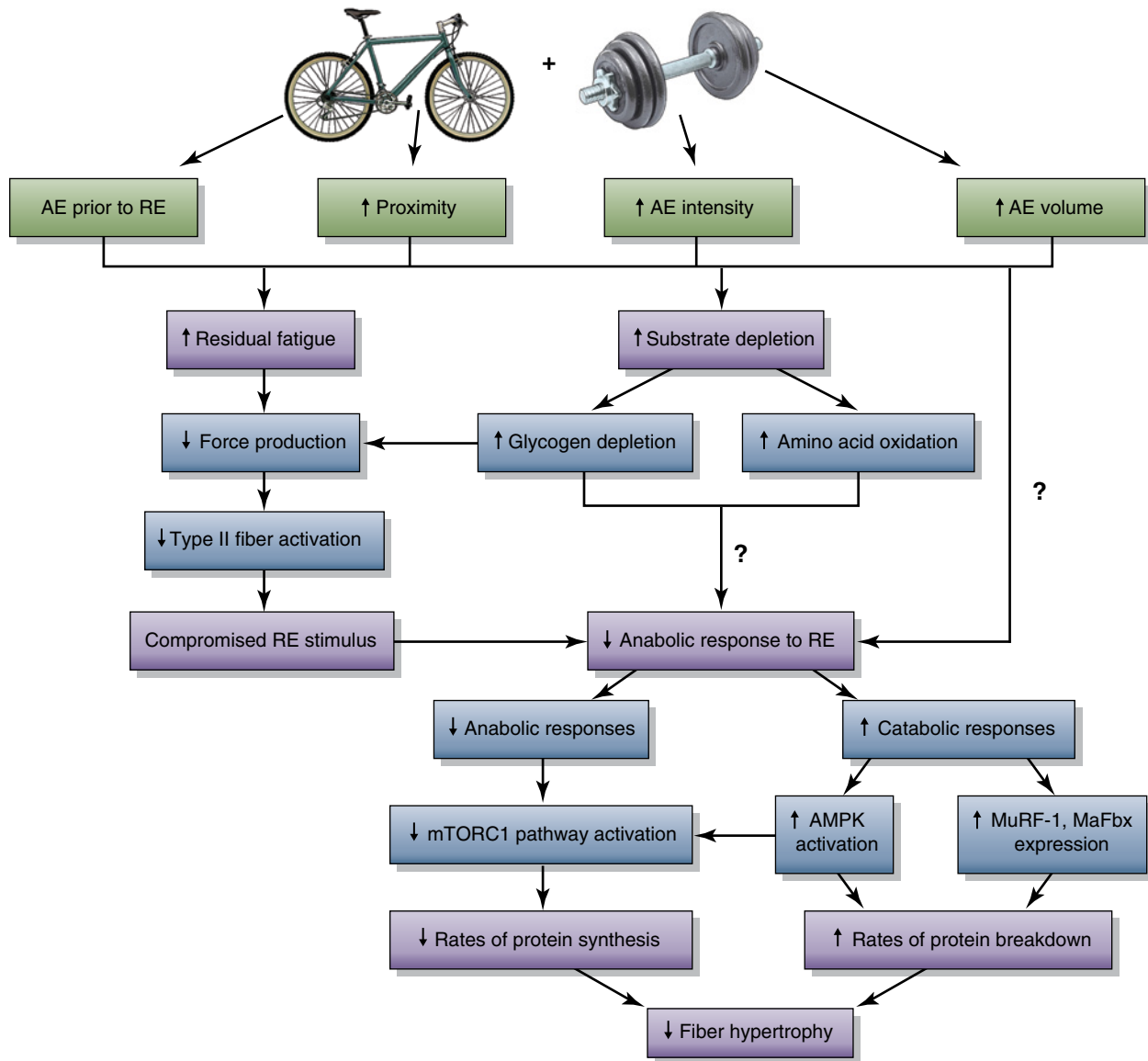


FIGURE 6.2 Chronic interference hypothesis. AE = aerobic exercise; RE = resistance exercise.

Reprinted from J.J. Fyfe, D.J. Bishop, and N.K. Stepto, "Interference Between Concurrent Resistance and Endurance Exercise: Molecular Bases and the Role of Individual Training Variables," *Sports Medicine* 44, no. 6 (2014): 743-762, with kind permission from Springer Science + Business Media.

is to maximize hypertrophy during concurrent training. However, caution must be used when extrapolating conclusions from non-matched studies and isolated intracellular signaling data, particularly given the general lack of correlation between acute molecular events and chronic hypertrophy in untrained subjects (2).

Long-term studies on muscular adaptations associated with varying aerobic intensities are similarly scarce. Silva and colleagues (65)

randomly assigned 44 young women to one of four groups:

1. Concurrent resistance and continuous running training
2. Concurrent resistance and interval running training
3. Concurrent resistance and continuous cycle ergometer training
4. Resistance training only

Results showed that all groups significantly increased measures of maximal strength and local muscular endurance, with no differences observed between the groups. Muscle hypertrophy was not assessed, however, precluding conclusions about the effects of intensity on growth.

Fyfe and colleagues (32) randomized untrained young men to perform either resistance training only, resistance training plus moderate-intensity aerobic exercise, or resistance training plus high-intensity interval training for 8 weeks. Results showed that only the group performing resistance training alone increased Type I fiber area; an atrophic effect was observed for both concurrent training groups. Interestingly, the group performing a combination of resistance training and interval training displayed the greatest decrease in Type I fiber size. No between-group differences were seen in Type II fiber hypertrophy. Overall, current evidence suggests a potential detriment to concurrent training on muscle growth. However, the paucity of research on the topic makes it difficult to draw definitive conclusions on the topic.

Volume and Frequency

Volume may have the biggest impact on the hypertrophic interference associated with concurrent training, potentially related to overtraining symptoms induced by a catabolic hormonal environment and chronic muscle glycogen depletion (58). This contention is supported by research showing attenuations in maximal strength with frequencies of more than 3 sessions per week but not less than 2 sessions per week (31). Pooled data from Wilson and colleagues (79) revealed a significant negative correlation between muscle hypertrophy and the volume (duration and frequency) of aerobic exercise during concurrent training. With respect to the specific components of volume, inverse correlations were especially strong for the duration of exercise ($r = .75$), whereas frequency displayed a relatively weak correlation ($r = .26$).

The effect of varying aerobic frequencies on muscular adaptations was directly studied in the context of a concurrent training program

KEY POINT

If hypertrophy is the desired outcome, the frequency of aerobic endurance training should remain low and a lengthy intervening recovery period should be inserted between aerobic and resistance bouts. Perhaps even better, the two should be performed on separate days.

(46). Subjects performed a 3-day-a-week resistance protocol and supplemented it with 0, 1, or 3 days of aerobic endurance training. Results showed an inverse dose–response relationship between increases in limb girth and aerobic frequency (4.3%, 2.8%, and 1% for the 0-, 1-, and 3-day-a-week conditions). These findings indicate that the frequency of aerobic endurance training should remain low if muscle hypertrophy is the primary desired outcome.

Mode

Although aerobic exercise can be carried out using a variety of modalities, running and cycling have primarily been studied in the context of concurrent training. The meta-analysis by Wilson and colleagues (79) revealed that running had a particularly negative effect on the hypertrophic adaptations associated with resistance training, whereas cycling did not appear to cause a significant detriment. The authors speculated that running-related impairments on muscle growth could be related to excessive muscle damage caused by its high eccentric component. Conceivably, this could inhibit recuperative abilities and thus blunt the post-exercise adaptive response. Alternatively, they proposed that cycling has greater biomechanical similarities to multi-joint free weight exercise compared to running and therefore may have provided a greater transfer of training. Counterintuitively, Panissa and colleagues (61) reported that high-intensity aerobic cycling negatively affected strength to a greater degree than high-intensity treadmill running when performed immediately before a resistance training bout. Over time, this would likely have a detrimental impact on hypertrophy as

a result of chronic reductions in mechanical tension. Overall, the evidence remains unclear as to whether a given aerobic modality interferes more with muscular adaptations when performed in combination with a regimented resistance training program; under certain circumstances, both running and cycling may have deleterious effects.

Scheduling

Depending on the scope of the training program, aerobic endurance exercise can be performed either in the same session with resistance training or on alternate days. Several studies have examined how the order of aerobic and resistance exercise performed in the same session affects intracellular signaling responses. Researchers have put forth the acute interference hypothesis whereby performing aerobic training immediately before strength training produces residual fatigue that ultimately compromises force production during resistance exercise (45). However, the hypothesis is specific to strength adaptations and does not necessarily reflect resultant effects on muscle hypertrophy despite a logical rationale that a reduction in mechanical tension would indeed compromise anabolism.

Coffey and colleagues (18) investigated the acute intracellular response to a combined session of knee extension resistance exercise and moderate-intensity cycling. Cycling before resistance exercise resulted in a heightened phosphorylation of Akt but a reduction in IGF-1 mRNA; alternatively, reversing the order of performance elevated concentrations of MuRF-1 mRNA. Follow-up work by the same lab revealed that performing a high-intensity sprint cycling bout prior to knee extensions blunted phosphorylation of p70^{S6K} compared to performing resistance exercise first (17). Moreover, the upregulation of translation initiation via the PI3K/Akt signaling pathway may be altered when resistance training is performed after glycogen-depleting aerobic exercise (21). Combined, these data suggest greater interference occurs when aerobic exercise precedes a resistance bout. That said, Babcock and colleagues (7) found that performing a 90-minute bout of aerobic cycling immediately

after resistance exercise completely blunted the post-workout satellite cell response seen with resistance training alone.

Data on the long-term effects of the order of same-day concurrent training on muscular adaptations are limited. Multiple studies show that strength gains are similar regardless of the sequence of training (14, 20, 35). Hence, mechanical tension does not appear to be compromised by the order of performance. From a hypertrophy standpoint, Cadore and colleagues (11) found similar increases in upper- and lower-body muscle thickness independent of whether aerobic or resistance training was performed first in a session. Similarly, Davitt and colleagues (22) found that changes in body composition were unaffected by aerobic endurance exercise either before or after resistance training. These studies seem to cast doubt on the importance of training sequence as a variable during concurrent training.

That said, the effects of order may be intensity dependent. Higher-intensity aerobic endurance exercise impedes subsequent force production, whereas lower-intensity continuous aerobic exercise tends to have less of an effect on residual fatigue (31). Both high-intensity cycling and treadmill exercise were shown to negatively affect the maximum number of repetitions and total session volume of a resistance training protocol performed after the aerobic bout (61). Interestingly, the extent of interference was highest after cycling compared to running. Moderate-intensity cycling performed subsequent to arm curl exercise was found to impair hypertrophy of the elbow flexors compared to arm curl exercise alone (71). Alternatively, other research shows no negative effects on hypertrophy when high-intensity interval cycling is performed after a heavy resistance exercise bout (73). Residual fatigue from previous aerobic training also negatively affects the volume of work performed during subsequent resistance training (31). Given the well-established dose–response relationship between volume and muscular adaptations, such reductions in total work may impede hypertrophy over time.

Taking the body of literature on the topic into account, interference appears to be

best minimized by either inserting a lengthy intervening recovery period between aerobic and resistance bouts or, perhaps even better, performing them on separate days. Indeed, Wilson and colleagues (79) found a trend for greater hypertrophy when aerobic and resistance exercise were performed on separate days as opposed to in the same session (effect size of 1.05 vs. 0.8, respectively).

Interestingly, performing an acute resistance training bout 6 hours after aerobic-oriented cycle ergometry was shown to elicit greater mTOR and p70^{S6K} phosphorylation compared to performing resistance training alone (53). This suggests that the aerobic bout actually potentiated anabolic signaling. The practical implications of these findings are undetermined.

RESEARCH FINDINGS

CONCURRENT TRAINING

Research indicates that concurrent training can have a negative impact on hypertrophic adaptations. Mitigating aerobic volume or intensity, or both, reduces the potential for negative consequences associated with the strategy. Non-weight-bearing aerobic activities such as cycling appear to attenuate deleterious effects compared to running, although some evidence is contradictory. There is an absence of research on the effects of cross-training on various modalities in the context of a regimented resistance training program. Whether such variation would enhance or hinder results remains speculative.

The majority of concurrent training studies have been carried out with untrained subjects, making it difficult to extrapolate conclusions to physically active people. The few studies that have employed subjects experienced in exercise training indicate greater interference in those who are well trained. Kraemer and colleagues (52) investigated the compatibility of aerobic and resistance exercise in a group of army recruits involved in standard military training for at least 3 days per week for 2 years before the onset of the study. Subjects were randomly assigned to perform aerobic endurance exercise, resistance exercise, or concurrent training. The aerobic endurance protocol consisted of a combination of steady-state and high-intensity interval training. After 12 weeks, subjects in the resistance-only group displayed increases in Type I, Type IIa, and Type IIc fiber diameters, whereas those in the concurrent group showed significant increases only in Type IIa fibers. Bell and colleagues (8) found similar results in a group of physically active university students, at least some of whom had experience in strength and aerobic endurance training. Subjects performed 12 weeks of cycle ergometry, resistance training, or a combination of both modalities. Results showed that resistance training only increased both Type I and Type II fiber cross-sectional area, whereas concurrent training produced increases only in Type II fibers. Moreover, the magnitude of Type II fiber hypertrophy was markedly greater in the resistance-only group compared to those who performed concurrent training (28% vs. 14%, respectively). Taken together, these findings suggest that concurrent training may be particularly detrimental to well-trained individuals.

Consideration also must be given to the relatively short duration of most concurrent training studies. Hickson (40) found no evidence of interference in a combined aerobic and resistance protocol until the 8th week of training. This finding indicates that negative effects on hypertrophy may not manifest for months, but ultimately long-term increases in muscle size may be compromised, conceivably as a result of nonfunctional overreaching/overtraining.

A case can be made that adding aerobic exercise to a resistance training program may indirectly help to augment long-term hypertrophy by improving blood flow capacity. It is well established that aerobic exercise increases angiogenesis via both vascular remodeling and increased capillarization (10). These adaptations have the potential to influence muscular adaptations in several ways. For one, enhanced flow to muscles allows for greater delivery of oxygen, growth factors, and macronutrients, which in turn may facilitate their ability to remodel. In addition, evidence indicates that individuals with higher capillary densities and a greater capacity for muscle perfusion display a heightened activation or expansion of the sat-

ellite cell pool, or both, conceivably increasing growth potential over time. Research indicates that resistance training alone is insufficient for increasing capillarization; only concurrent training enhances such adaptations (73). As discussed, however, the intensity and duration of aerobic training must be considered in conjunction with the volume of resistance training performed so that the negative effects of nonfunctional overreaching/overtraining do not override potential benefits of enhanced angiogenesis. Moreover, order of scheduling needs to be taken into account as well, with sufficient recovery intervals afforded between aerobic and resistance exercise bouts.

TAKE-HOME POINTS

- Aerobic exercise can promote increases in muscle hypertrophy in untrained people, and gains are primarily limited to Type I fibers. The extent of hypertrophic adaptations is contingent on intensity, volume, frequency, and mode of training, as well as the person's level of deconditioning.
- Aerobic intensities of >80% of HRR are generally required to promote gains in muscle mass in untrained people.
- Although highly deconditioned people can experience hypertrophic increases with relatively low volumes of aerobic training, those who are more active require higher training volumes.
- Evidence suggests that cycling exercise may be more conducive to increasing muscle mass than walking, running, or jogging, possibly because ambulatory activities are performed more often in daily life.
- Concurrent training can interfere with hypertrophic adaptations. Higher aerobic volumes appear particularly detrimental in this regard, although the effect of high aerobic intensities is not well elucidated.
- The negative effects of concurrent training are best minimized by either inserting a lengthy intervening recovery period between aerobic and resistance bouts or, perhaps even better, performing them on separate days.
- If properly structured, the addition of aerobic exercise to a resistance training program may facilitate better long-term hypertrophic increases via beneficial effects of angiogenesis.